ETHERNET FRAMES (PART 1)

1. ETHERNET ENCAPSULATION (7.1.1)

Ethernet technology includes explanation of MAC sublayer and Ethernet frame fields.

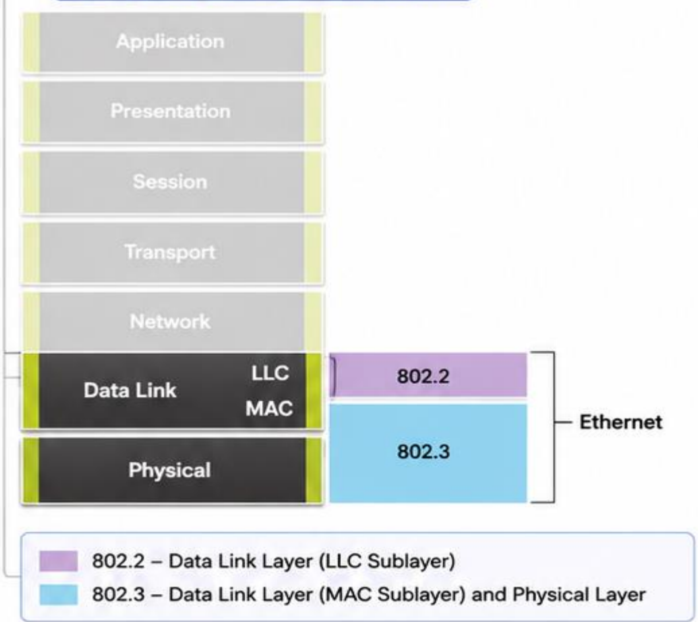
Ethernet is a LAN technology used today. Other LAN technologies: Wireless LANs (WLANs), Token Ring, fiber-optic links, and coaxial cable.

Ethernet operates in the data link layer and physical layer. It is defined in IEEE 802.2 (LLC) and 802.3 (MAC). Ethernet supports the following data bandwidths:

- 10 Mbps
- 100 Mbps
- 1000 Mbps (1 Gbps)
- 100,000 Mbps (100 Gbps)
- 1000 Mbps (1 Gbps)
- 10,000 Mbps (10 Gbps)
- 40,000 Mbps (40 Gbps)

Ethernet standards define both the Layer 2 (data link) protocol and the Layer 1 (physical) technologies.

ETHERNET AND THE OSI MODEL



KEY TAKEAWAY Ethernet uses 802.2 (LLC) and 802.3 (MAC + Physical). | Supports multiple data rates from 10 Mbps to 100 Gbps.

2. DATA LINK SUBLAYERS (7.1.2)

IEEE 802 LAN/MAN protocols, including Ethernet, use two sublayers in the data link layer:

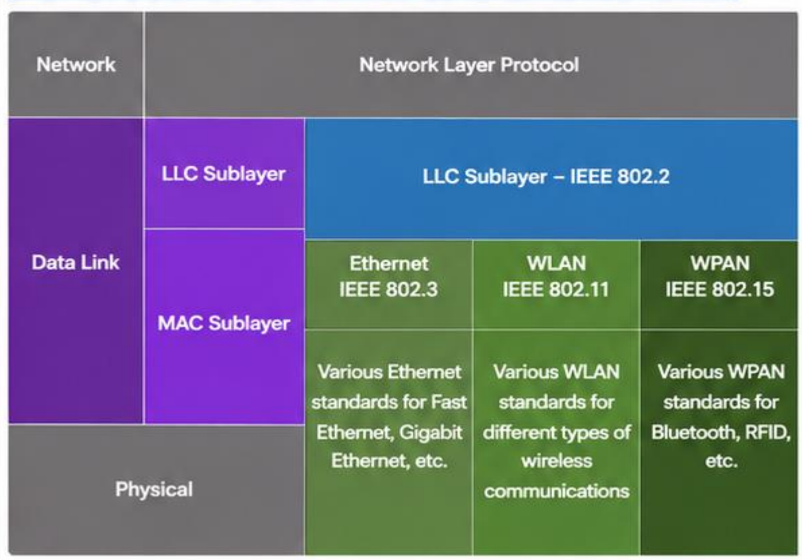
A. LLC SUBLAYER (Logical Link Control)

- IEEE 802.2 sublayer.
- Communicates between network software (upper layers) and device hardware (lower layers).
- Places information in the frame that identifies which network layer protocol is being used.
- Supports multiple Layer 3 protocols (e.g., IPv4, IPv6) on the same network interface and media.

B. MAC SUBLAYER (Media Access Control)

- Implemented in hardware.
- Responsible for data encapsulation and media access control.
- Provides data link layer addressing.
- Integrated with various physical layer technologies.

DATA LINK LAYER SUBLAYERS AND NETWORK PROTOCOLS



KEY NOTES

- LLC sublayer handles the network protocol data and identifies the Layer 3 protocol.
- MAC sublayer handles addressing, encapsulation, media access control, and error detection.

ETHERNET FRAMES (PART 2)

1. MAC SUBLAYER (7.1.3)

The MAC sublayer is responsible for data **encapsulation** and **accessing** the media.

A. DATA ENCAPSULATION (IEEE 802.3)

IEEE 802.3 data encapsulation includes:

- **Ethernet frame** – Internal structure of the Ethernet frame.
- **Ethernet addressing** – Adds source and destination MAC addresses to deliver the frame from source NIC to destination NIC on the same LAN.
- **Ethernet error detection** – Adds a Frame Check Sequence (FCS) trailer for error detection.

B. ACCESSING THE MEDIA

IEEE 802.3 MAC sublayer defines how Ethernet works over different media types, including copper and fiber.

ETHERNET STANDARDS IN THE MAC SUBLAYER

Network	Network Layered Protocol					
Data Link	LLC Sublayer	LLC Sublayer – IEEE 802.2				
	MAC Sublayer	Ethernet – IEEE 802.3				
		802.3u Fast Ethernet	802.3z Gigabit Ethernet over Fiber	802.3ab Gigabit Ethernet over Copper	802.3ae 10 Gigabit Ethernet over Fiber	Etc.
Physical						

LEGACY ETHERNET (BUS / HUBS)

A shared, half-duplex medium.
Uses contention-based CSMA/CD.
Collision Detection (CD) part of CSMA/CD.
Provides a back-off algorithm for retransmission.

SWITCHED ETHERNET

Full-duplex communication.
Does not require CSMA/CD.
Switches provide dedicated links, eliminating collisions.

KEY TAKEAWAY

- MAC sublayer handles framing, addressing, and error detection.
- It ensures reliable delivery of frames on the local media.

2. ETHERNET FRAME FIELDS (7.1.4)

IMPORTANT POINTS

- Minimum Ethernet frame size = 64 bytes.
- Maximum Ethernet frame size = 1518 bytes. (Includes all bytes from Destination MAC to FCS field. Preamble not included.)
- VLAN tagging (if used) is beyond the scope of this course.
- Frames < 64 bytes are considered "collision fragments" or "runt frames" and are discarded.
- Frames > 1518 bytes are "jumbo" or "baby giant" frames.
- Dropped frames are invalid or unsupported (by most Fast Ethernet, Gigabit Ethernet switches and NICs).
- Field sizes are in transmitted order.

ETHERNET FRAME FORMAT (64 – 1518 BYTES)

8 bytes	6 bytes	6 bytes	2 bytes	46 – 1500 bytes	4 bytes
Preamble and SFD	Destination MAC Address	Source MAC Address	Type / Length	Data	FCS (Frame Check Sequence)

NOTES

- All sizes are in bytes.
- Preamble and SFD are transmitted on the wire but are not counted in frame size.
- Data field size is variable: 46 – 1500 bytes.

ETHERNET FRAME FIELDS – DETAILS

Field	Size	Description
Preamble and Start Frame Delimiter (SFD)	8 bytes (7 + 1)	Preamble (7 bytes) and Start Frame Delimiter (SFD, 1 byte). Helps receiving devices synchronize with the sender and recognize the start of a new frame.
Destination MAC Address	6 bytes	Identifies the intended recipient. Used by Layer 2 to decide if the frame is for this device. If address ≠ device's MAC, the frame is discarded. Can be unicast, multicast, or broadcast.
Source MAC Address	6 bytes	Identifies the originating NIC or interface that sent the frame.
Type / Length	2 bytes	Identifies the upper layer protocol encapsulated in the frame. Common values: 0x0800 = IPv4, 0x86DD = IPv6. **Note:** You may also see this field referred to as EtherType, Type, or Length.
Data Field	46 – 1500 bytes	Contains the encapsulated data from a higher layer (e.g., Layer 3 PDU like IPv4/IPv6 packet). Minimum size = 46 bytes. If data is smaller, padding is added to meet the minimum size.
Frame Check Sequence (FCS)	4 bytes	CRC (Cyclic Redundancy Check) value used for error detection. Sender calculates CRC and places it in this field. Receiver recalculates CRC; if it doesn't match, the frame is dropped (indicating an error or corruption during transmission).

QUICK SUMMARY

- **MAC sublayer:** encapsulation, addressing, and error detection.
- **IEEE 802.3** defines Ethernet in the data link & physical layers.
- **Frame size:** 64 – 1518 bytes (Dest MAC to FCS).
- FCS ensures reliable delivery by detecting errors.
- CSMA/CD used in shared media; not needed in full-duplex switched Ethernet.

ETHERNET MAC ADDRESS (PART 1)

1. MAC ADDRESS AND HEXADECIMAL (7.2.1)

Understanding MAC addresses requires converting between decimal numbers, hexadecimal numbers, and binary numbers.

An example MAC address used in this course is:
00-0C-85-69-96-10

MAC addresses are 48 bits **(6 bytes)** long and written in hexadecimal.

Among the various numbering systems, hexadecimal is the most common for MAC addresses.

The table on the right shows hexadecimal values and their equivalent decimal and binary values.

CONVERTING AND COUNTING BY 16s IN HEXADECIMAL

Decimal	Hex (Base 16)	Binary (4 bits)
0	0000	0000
1	0001	0001
2	0010	0010
3	0011	0011
4	0100	0100
5	0101	0101
6	0110	0110
7	0111	0111
8	1000	1000
9	1001	1001
10	1010	1010
11	1011	1011
12	1100	1100
13	1101	1101
14	1110	1110
15	1111	1111

KEY POINT

- Each MAC address byte can have a value from **00 to FF** in hexadecimal.
- Therefore, a full MAC address has **6 bytes (12 hexadecimal digits)** separated by hyphens.

2. BINARY, DECIMAL, AND HEX CONVERSIONS (EXAMPLES)

BINARY, DECIMAL (BASE 10), AND HEXADECIMAL (BASE 16) CONVERSIONS

Decimal	Binary	Hexadecimal
0	0000 0000	00
1	0000 0001	01
2	0000 0010	02
3	0000 0011	03
4	0000 0100	04
5	0000 0101	05
6	0000 0110	06
7	0000 0111	07
8	0000 1000	08
9	0000 1001	09
10	0000 1010	0A
11	0000 1011	0B
12	0000 1100	0C
13	0000 1101	0D
14	0000 1110	0E
15	0000 1111	0F

EXAMPLES

- Each hexadecimal digit represents 4 binary bits.
- Each byte = 8 bits = 2 hexadecimal digits.
- Example: Hex 0C = Binary 0000 1100
- Example: Hex 85 = Binary 1000 0101
- Example MAC address:
  00-0C-85-69-96-10

IMPORTANT NOTES

- The binary representation is of length 48, which is equivalent to 6 bytes (8 bits per byte).
- The MAC address is written in hexadecimal because it is shorter and easier to read.
- This notation is the standard used in networking and device identification.

IN SUMMARY

- MAC Address length = 48 bits = 6 bytes = 12 hexadecimal digits.
- Hexadecimal notation (00 to FF) is commonly used to represent each byte.
- Binary, Decimal, and Hexadecimal conversions are essential for understanding and troubleshooting networks.

ETHERNET MAC ADDRESS (PART 2)

1. ETHERNET MAC ADDRESS (7.2.2)

In an Ethernet LAN, every device is connected to the same, shared media.

The MAC address identifies the **physical source and destination devices** (NICs) on the local network segment.

An **Ethernet MAC address** is **48 bits** long (6 bytes) expressed using **12 hexadecimal digits** (separated by hyphens).

All MAC addresses must be **unique** on the Ethernet device or Ethernet interface.

IEEE assigns vendors a **unique 6-hexadecimal** (24-bit or 3-byte) code called the **Organizationally Unique Identifier (OUI).**

When a vendor assigns a MAC address:

- Use the assigned **OUI** as the first 6 hex digits.
- Assign a **unique** value in the **last 6 hex digits.**

DECIMAL, BINARY, AND HEXADECIMAL EQUIVALENTS (0 TO F)

Decimal	Binary (4 bits)	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

KEY POINT

- **First 6 hex digits** = Vendor's OUI (24 bits).
- **Last 6 hex digits** = Vendor-assigned unique value (24 bits).
- **Full MAC address = 6 bytes = 12 hexadecimal digits.**

MAC ADDRESS FORMAT (48 BITS = 6 BYTES)

Organizationally Unique Identifier (OUI) – First 24 bits (3 bytes)			Vendor Assigned – Last 24 bits (3 bytes)		
1111 1111	1111 1111	1111 1111	1111 1111	1111 1111	1111 1111
8 bits	8 bits	8 bits	8 bits	8 bits	8 bits
Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6

= 48 bits

= 12 hex digits

= 6 bytes

EXAMPLE

- Assume Cisco is assigned OUI: **00-00-2F**
- Cisco configures a unique vendor code: **3A-07-BC**
- Therefore, the MAC address of the device is:

00-00-2F-3A-07-BC

NOTE

- It is the vendor's responsibility to ensure that none of its devices be assigned the same MAC address.
- Duplicate MAC addresses can occur due to manufacturing mistakes, virtualization, or improper configuration.
- Tools and processes are used to detect and resolve duplicates.

2. FRAME PROCESSING (7.2.3)

The MAC address in the frame is the **Burned-In Address** (BIA) because it is hard coded into read-only memory (**ROM**) on the NIC. This means the **address is encoded into the ROM** chip permanently.

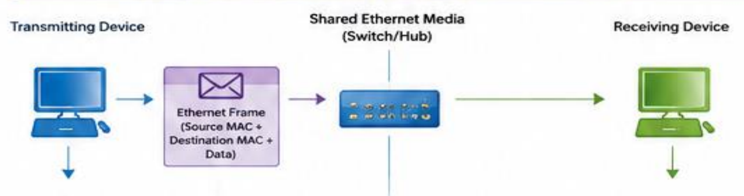
The OS uses the operating system and NIC to access the local network in software. It is used when attempting to gain access to a network that restricts based on MAC.

When the computer boots up, the NIC copies its MAC **address from ROM into RAM.**

When a device needs to transmit a message on an Ethernet network, the frame must include:

- **Source MAC address** – MAC of the source device NIC.
- **Destination MAC address** – MAC of the destination device NIC.

HOW A FRAME IS PROCESSED (FRAME FORWARDING)



1. Device creates frame with Source and Destination MAC addresses.
2. Frame is sent over the Ethernet media (switch/hub).
3. Destination device NIC receives the frame and checks the Destination MAC address.
4. If MAC matches its own, the frame is accepted and passed up to the OS layers, where it is processed.

Click Play in the animation to view the frame forwarding process.

IMPORTANT NOTES

- Ethernet NICs will accept frames if the destination MAC address is:
  - A broadcast address
  - A multicast address
- Any device that is the source or destination of an Ethernet frame will have an Ethernet NIC and therefore a MAC address.
- Examples: workstations, servers, printers, mobile devices, routers.
- Ethernet FCS will also accept frames if the destination MAC address is a broadcast or a multisost group address.

ETHERNET MAC ADDRESS (PART 3)

1. UNICAST MAC ADDRESS (7.2.4)

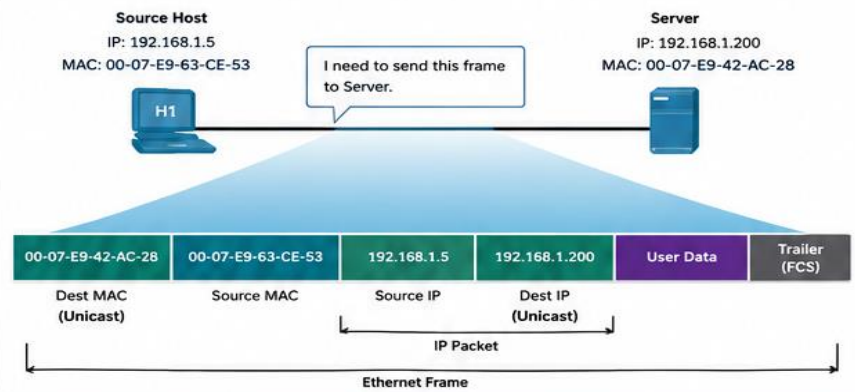
Different MAC addresses are used for Layer 2 unicast, broadcast, and multicast communications.

A unicast MAC address is the unique address used to send a frame from one transmitting device to a single destination device.

MAC addresses are 48 bits (6 bytes) long written in hexadecimal.

In the example, the destination MAC and the destination IP address are both unicast.

EXAMPLE: UNICAST FRAME



KEY POINTS

- The destination IP address in the IP packet header must be a unicast address.
- The destination MAC address in the Ethernet frame header must also be a unicast address.
- Source MAC address must always be a unicast address.

NOTE

A source host uses ARP (Address Resolution Protocol) to find the destination MAC address associated with the destination IP address.

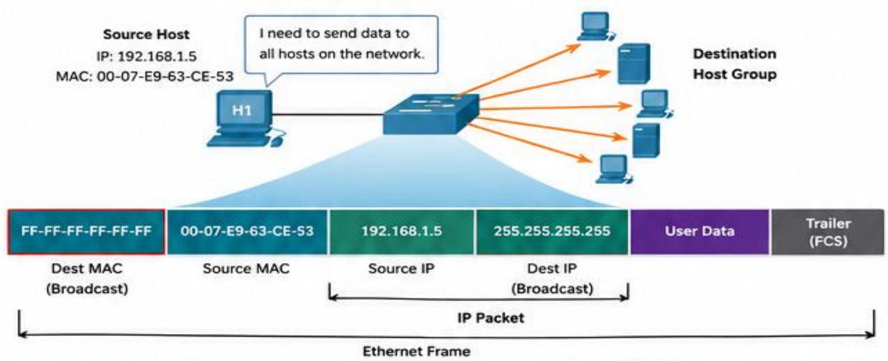
2. BROADCAST MAC ADDRESS (7.2.5)

BROADCAST FEATURES

- Destination MAC address: FF-FF-FF-FF-FF-FF (48 ones in binary).
- Flooded out all Ethernet switch ports except the incoming port.
- Not forwarded by a router.
- Hosts on the local network (broadcast domain) will receive and process it.

If the encapsulated data is an IPv4 broadcast packet, the destination IP address in the IP header is 255.255.255.255 (all 1s).

EXAMPLE: IPV4 BROADCAST FRAME



KEY POINTS

- Destination MAC = FF-FF-FF-FF-FF-FF
- Destination IP (IPv4 broadcast) = 255.255.255.255
- Used to reach all devices on the local broadcast domain.

EXAMPLE SCENARIO

A host needs to reach all devices on the network, such as sending discovery requests.

NOTE

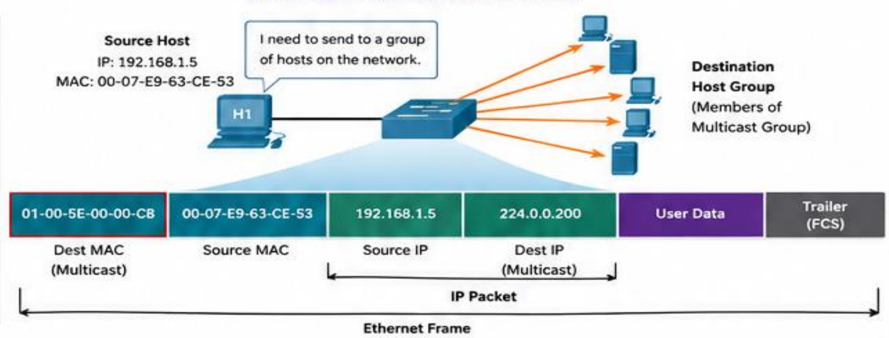
DHCP for IPv4 uses broadcast frames to discover DHCP servers.
ARP Requests are also sent as broadcast frames.

3. MULTICAST MAC ADDRESS (7.2.6)

MULTICAST FEATURES

- Used to send a frame to a group of devices that belong to the same multicast group.
- Destination MAC address range: 01-00-5E-00-00-00 to 01-00-5E-7F-FF-FF
- Not reserved for Spanning Tree Protocol (STP) and Link Layer Discovery Protocol (LLDP).
- Flooded out all ports except the incoming port, unless the switch is configured for multicast snooping.
- Not forwarded by a router, unless the router is configured to route multicast packets.

EXAMPLE: IPV4 MULTICAST FRAME



KEY POINTS

- Multicast MAC range: 01-00-5E-00-00-00 to 01-00-5E-7F-FF-FF
- Destination IP (multicast) range: 224.0.0.0 to 239.255.255.255
- Deliver frames only to members of the multicast group.

USE CASES

Common in streaming media, video conferencing, IPTV, stock tickers, routing protocols, etc.

NOTE

Routing protocols work atther network protocols use multicast addressing.

QUICK SUMMARY

UNICAST
One-to-one communication.
Unique destination MAC and IP addresses.

BROADCAST
One-to-all on local network.
MAC = FF-FF-FF-FF-FF-FF
IP = 255.255.255.255

MULTICAST
One-to-many (group members).
MAC = 01-00-5E-xx-xx-xx
IP = 224.0.0.0 – 239.255.255.255

REMEMBER
MAC addresses work at Layer 2.
Use the right type to deliver frames efficiently and securely.